



ARTIFICIAL INTELLIGENCE

The Unsung Eco-Warriors of the AI Age

Why the people squeezing bits out of neural networks might be the climate movement's most unlikely allies.

By **DHEERAJ**

A report on AI's quiet efficiency revolution

PICTURE A CLIMATE ACTIVIST. You probably pictured someone outdoors. Banner, megaphone, maybe a kayak alongside an oil tanker. You probably did not picture an ML researcher, third coffee deep, squinting at perplexity curves and trying to convince themselves that 2.5 bits per weight is enough to preserve a 70-billion parameter model's grasp on basic arithmetic.

This is a piece in defense of an obscure tribe of engineers — the people who work on *quantization*, the art of representing the brains of AI models in fewer bits — and the argument that their ungla-morous, deeply mathematical work may be one of the more underappreciated climate-adjacent interventions happening inside tech.

I. AI'S POWER PROBLEM IS NOW IMPOSSIBLE TO IGNORE

The numbers are no longer ambiguous. In April 2026, the International Energy Agency reported that global data centre electricity demand grew 17% in 2025, and that AI-focused data centres grew at **50%** — well outpacing the 3% growth rate of global electricity demand overall. The IEA's Base Case projects data centre consumption to roughly double from 485 TWh in 2025 to 945 TWh by 2030, which would put data centres at around 3% of global electricity — close to the entire current consumption of Japan.

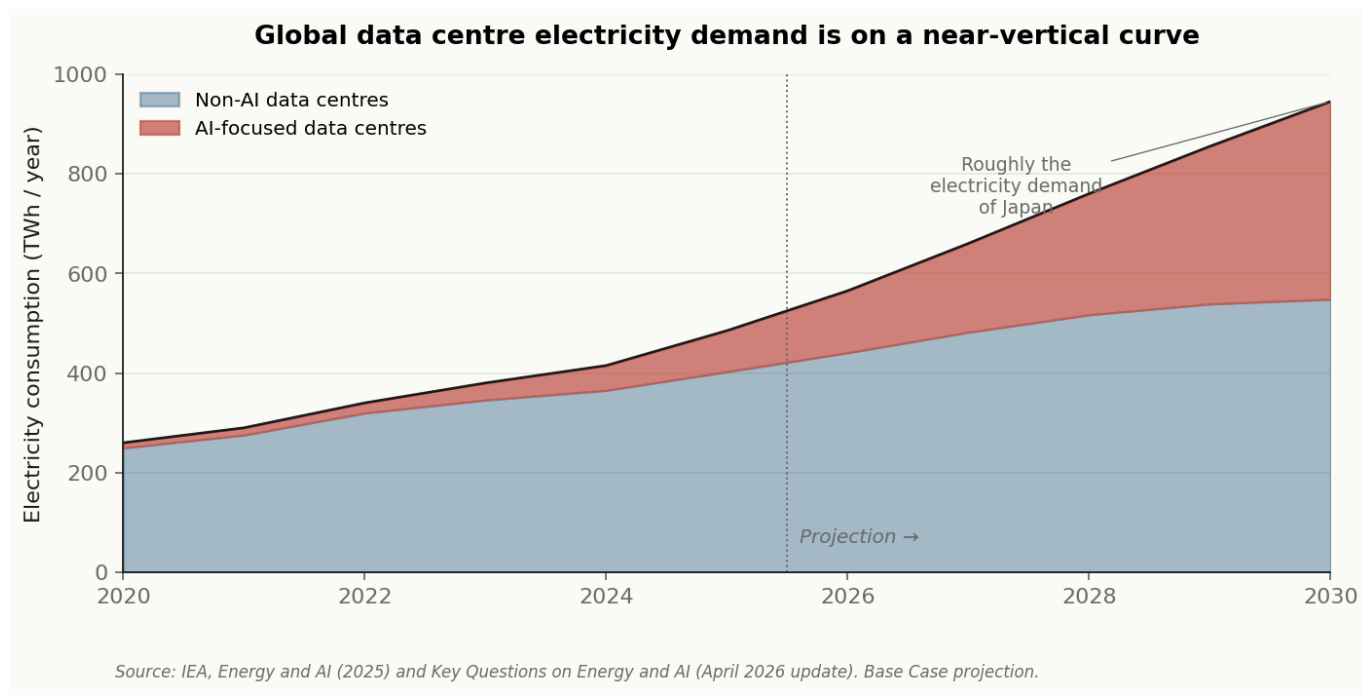


FIG. 01 Global data centre electricity demand, 2020–2030. The AI-focused share grows from a sliver in 2020 to roughly 40% of all data centre consumption by 2030. Source: IEA Base Case projection, April 2026 update.

That curve is being financed in real time: the five largest tech companies spent over **\$400 billion** in capital expenditure in 2025 on data centre build-out, and the IEA expects that figure to grow an-

other 75% in 2026. Hyperscalers are bidding on nuclear power purchase agreements. Long-dormant nuclear assets are being reconsidered as power sources for AI infrastructure. Data centre

operators are standing up on-site gas turbines because grid interconnects have multi-year waiting lists. New AI campuses are being routed around

whatever counties can supply hundreds of megawatts on short notice.

"The bottleneck on frontier AI deployment, today, is not chips or capital or talent. It is electrons."

— ON THE NEW PHYSICS OF AI SCALE

II. WHERE THE ENERGY ACTUALLY GOES

Inside an AI model, a surprising amount of energy is not spent on the arithmetic itself, but on re-

peatedly moving billions of numbers from memory to compute units. Inference looks like computation from the outside, but at the hardware level, much of the cost is data movement.

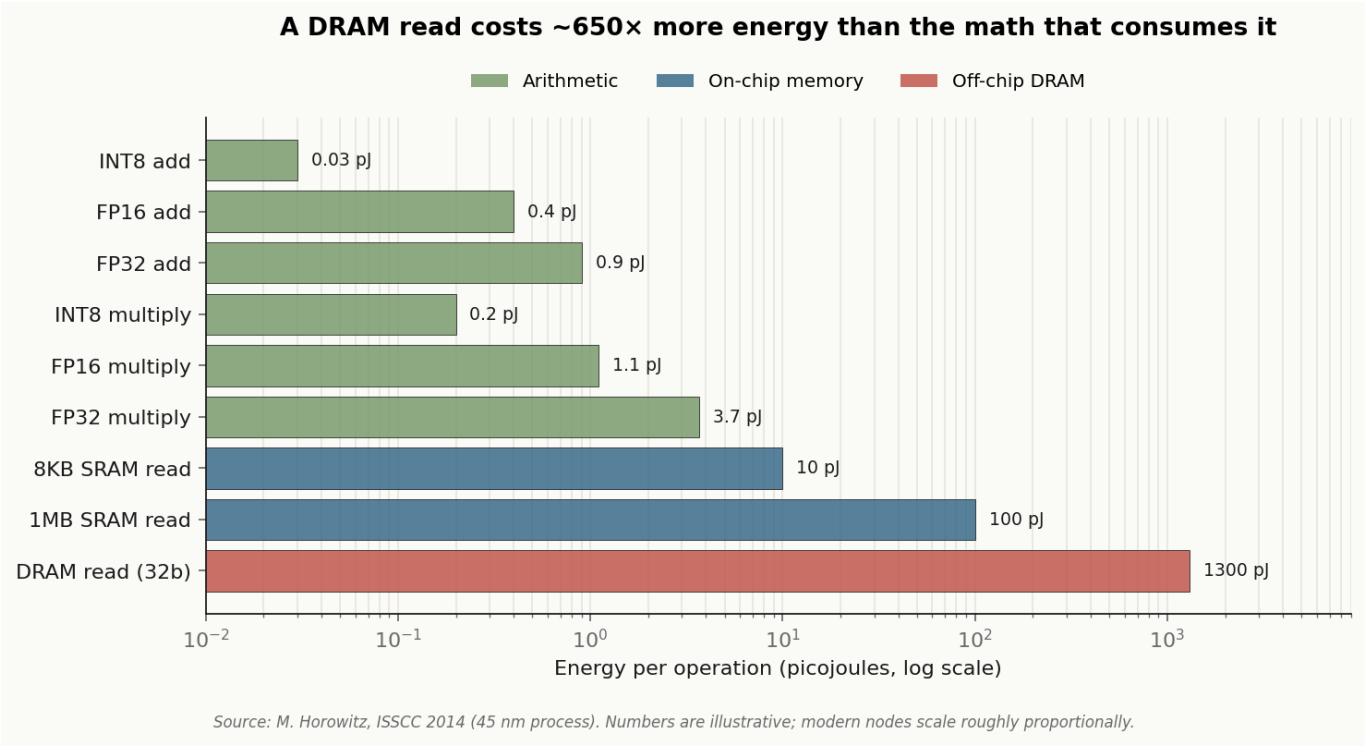


FIG. 02 Energy per operation at 45 nm. Reading a 32-bit value from off-chip DRAM costs roughly 650× more energy than the floating-point multiply that consumes it. Source: M. Horowitz, ISSCC 2014.

Reading a single 32-bit number from off-chip DRAM costs around 1,300 picojoules. The floating-point multiply that actually consumes it costs around 3.7 — roughly **350 to 650× less**, depending on the precision used. Reading from on-chip cache

is much cheaper, but still 25–100× more expensive than the math. Every time a language model generates a single word, it streams its entire memory of weights — billions of numbers — out of DRAM, across the chip, and into the compute units. The

arithmetic is the cheap part. The schlep is everything.

This is where the quantization engineers come in.

III. WHAT THEY ACTUALLY DO

A neural network, in the form it's trained, stores each of its weights as a 16-bit floating-point number. A quantization engineer's job is to figure out how to represent that same weight in 8 bits. Or 4. Or 3. Or, lately, in some cases, fewer than 2. With clever math (calibration techniques, sensitivity analyses, fancy lattice geometries, and other methods that mostly live in papers with terrifyingly spe-

cific titles) they preserve almost all of the model's intelligence while shrinking the bag of numbers it has to lug around to four or eight times smaller.

Cut the data you move by 4×, and you cut the energy of moving it by roughly 4×. And it shows up in the measurements that matter for data centres. An April 2026 benchmark from AIMultiple ran Qwen3-32B on a single NVIDIA H100 SXM — the workhorse GPU of the current AI build-out — at four different precision levels, using the vLLM serving framework. The throughput differences translate directly into energy differences, because an H100 at full inference load draws roughly the same power regardless of the format it's running:

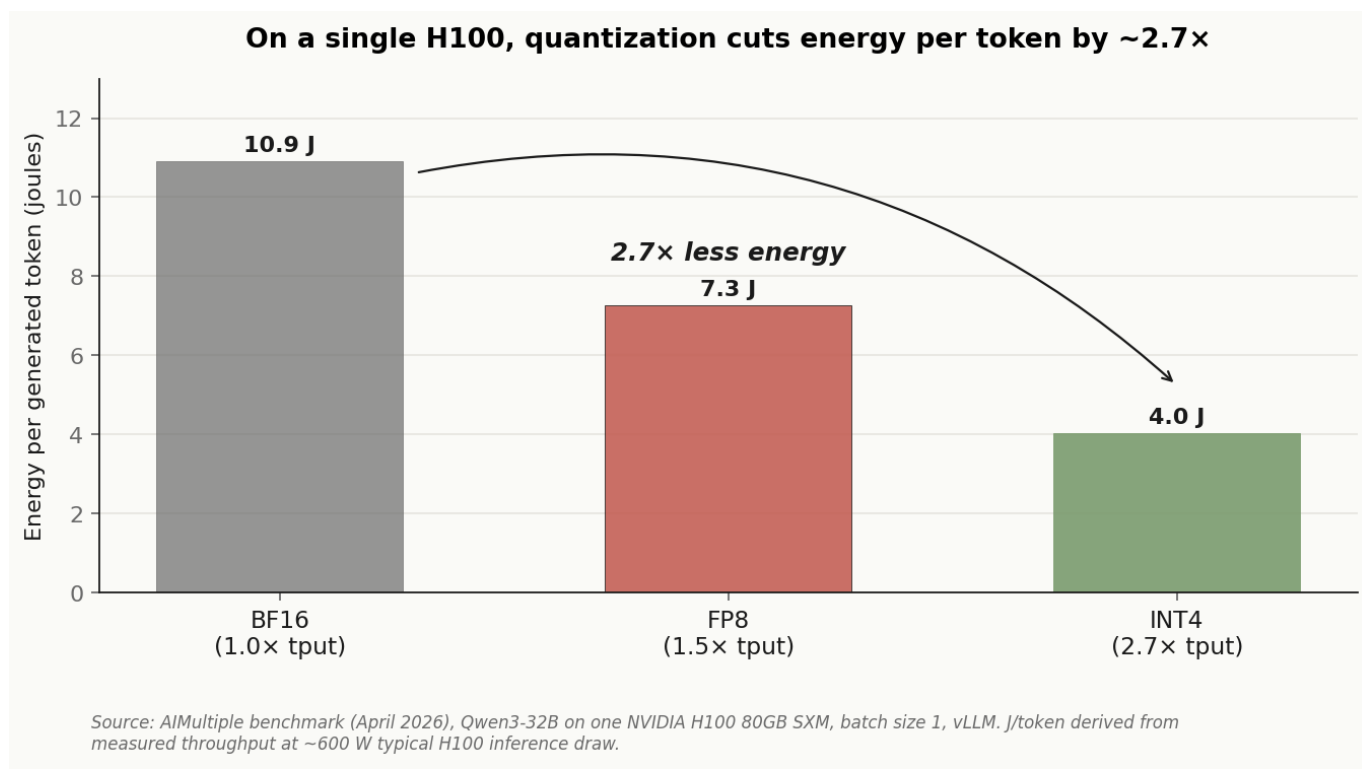


FIG. 03 Energy per generated token on a single NVIDIA H100 80GB SXM, running Qwen3-32B at three precision levels. BF16 to INT4 cuts energy per token by roughly 2.7×, the same factor as the throughput gain. Source: AIMultiple benchmark, April 2026.

A BF16 deployment burned about **10.9 joules per generated token**. The FP8 version of the same model — Nvidia's native 8-bit floating-point format, lossless across all reported benchmarks — cut that to **7.3 J**. The INT4 version came in at **4.0 J**

per token, a 2.7× reduction from BF16, with single-digit accuracy drops on the hardest reasoning benchmarks. Those gains compound: the same Spheron analysis found that FP8 roughly halves cost per token on H100-class hardware by doubling

throughput without adding a single GPU. And the December 2025 TokenPowerBench measurements on 16-GPU clusters running Llama-3 405B show the same trend at frontier scale — FP8 raises end-to-end throughput by about a third over FP16 with no measurable accuracy loss.

The IEA's own April 2026 report puts it bluntly: "power consumption per AI task is declining rapidly, with efficiency improving at a rate unprecedented in energy history." A non-trivial fraction of that efficiency improvement is happening in the quantization literature.

IV. THE COMPOUNDING NOBODY SEES

Here is the part that should make any environmentalist sit up: the savings *compound*. A climate hero putting solar on a roof or improving a building's efficiency creates real, continuous benefits — but those benefits are tied to where the work is physically deployed. Quantization works differently. Once a technique makes its way into an inference kernel or model-serving stack, it can travel across deployments, models, and every token generated by systems that adopt it. The work is highly reusable, and that is where the leverage comes from.

A small community of researchers, working on a problem that rarely gets public attention, end up multiplying their efficiency gains across the entire global inference fleet. It's not glamorous. There are no documentaries. The papers have titles like *Rotation-Invariant Mixed-Precision Calibration via Diagonal Fisher Approximations*. ***Nobody is going to put that on a t-shirt.***

But the carbon math is real.

V. THE ARGUMENT THAT ALWAYS COMES UP

Anyone who's spent five minutes thinking about efficiency and the environment will, at this point, raise the Jevons paradox. If you make AI cheaper to run, companies will just deploy more of it, and total energy will stay the same or grow. Efficiency is a discount on usage, not a guarantee of lower total consumption.

It's a fair concern, and it's been the rebuttal to every efficiency intervention since the steam engine. But it doesn't kill the case for quantization. It complicates it.

Consider where AI deployment is actually constrained right now. It isn't constrained by demand. It is constrained by *power*. When the binding limit on a data centre is the number of megawatts the grid can deliver, as it visibly is for the frontier players right now, efficiency doesn't just unlock more usage at the same energy. It unlocks more *useful output* from a fixed energy ceiling that already exists. The choice the planet faces isn't between "small efficient AI" and "no AI." It's between "small efficient AI inside a 500 MW campus" and "large wasteful AI demanding a 2 GW campus." Quantization is what keeps that second sentence from being the future.

There's also the counterfactual that almost never gets named. Whatever amount of AI ends up running in 2030, whatever the demand curve does, whatever the policy environment looks like: it is strictly better, for emissions, that it runs at lower joules per token than higher. You don't have to defeat the rebound argument to defend efficiency. You only have to want the realized workload, whatever it is, to use less energy than its inefficient alternative. That's a free win, every time.

And quantization opens a category of deployment that doesn't even *touch* a data centre: AI on

phones, on laptops, on cars, on cheap hardware in places without good connectivity. That's not rebound. That's a different game.

VI. WHERE THIS LEAVES US

I'm not going to argue that quantization engineers are saving the planet single-handedly. They're not. The grid still has to decarbonize. The bigger fight is upstream, in how we generate power. But the question of how efficiently we use the power we have is also a climate question, and the people who shave 75% off the energy cost of a model's infer-

ence deserve to be counted in the same ledger as the people putting solar on a roof.

They look like ML researchers. They post benchmarks on Twitter. They are obsessed with perplexity scores that many never read.

But check the carbon math, and they are climate workers in everything but title. Eco-warriors in disguise. The ones who showed up to the planetary problem with a laptop and a Hessian matrix instead of a banner. ■

A NOTE TO THE READER

If you're a researcher working on post-training quantization, low-bit inference kernels, or any of the hundred adjacent problems: nobody is going to throw you a parade. But you should know what you're actually doing. You're prying joules loose at industrial scale. That counts.

SOURCES & FURTHER READING

1. **Key Questions on Energy and AI** — International Energy Agency, April 2026 update. Data on global data centre electricity demand, AI-focused growth rate, 2030 projections, and hyperscaler capital expenditure.
2. **Energy and AI** — International Energy Agency, 2025. Baseline IEA Base Case modelling.
3. **LLM Quantization: BF16 vs FP8 vs INT4** — AIMultiple, April 2026. Direct benchmark of Qwen3-32B on a single NVIDIA H100 80GB SXM across BF16, FP8, INT8, and INT4 precision; throughput, accuracy, and cost-per-token measurements.
4. **TokenPowerBench: Benchmarking the Power Consumption of LLM Inference** — Niu et al., arXiv 2512.03024 (December 2025). Energy-per-token measurements at frontier scale, including Llama-3 405B on 16× H100 clusters.
5. **GPU Cost Per Token: Benchmarking 7 Major LLMs Across GPU Types** — Spheron Network, April 2026. Cluster-level analysis of FP16, FP8, INT4, and FP4 deployments on H100, H200, and B200.
6. **Computing's Energy Problem (and What We Can Do About It)** — M. Horowitz, ISSCC 2014. Per-operation energy figures for arithmetic and memory at 45 nm.